Development of the Hybrid MINBEE-TOPUS High-Resolution Scheme

1 Introduction

The development of high-resolution schemes requires a delicate balance between preserving sharp discontinuities without artificial oscillations (TVD constraints) and maintaining high accuracy in smooth regions.

The **MINBEE** scheme successfully bridges the highly compressive Superbee limiter and the highly diffusive Minmod limiter. It acts as a "tamed Superbee," preventing artificial stair-stepping on smooth shapes while keeping shocks sharp. However, as a strictly second-order TVD scheme, it naturally clips extreme high-frequency waves. Conversely, the **TOPUS** (Third-Order Polynomial Upwind Scheme) developed by Candezano, excels in strictly smooth, continuous domains by using polynomial formulations, but its non-linear damping makes it overly diffusive when subjected to severe discontinuities.

To overcome these individual limitations, we formulate a **Hybrid MINBEE-TOPUS Scheme** in the Normalized Variable Diagram (NVD) framework.

2 The Hybrid Flux Formulation

We define the new normalized face value, $\tilde{\phi}_{f,\text{Hybrid}}$, as a dynamic convex combination of the MINBEE and TOPUS schemes:

$$\tilde{\phi}_{f,\text{Hybrid}} = (1-\omega)\tilde{\phi}_{f,\text{MINBEE}} + \omega\tilde{\phi}_{f,\text{TOPUS}} \tag{1}$$

Where ω is a continuous smoothness sensor ($0 \leq \omega \leq 1$) that dictates the local transition between the two schemes based on the successive gradient ratio r.

3 Limiter Equivalents and the Smoothness Sensor (ω)

In Sweby's flux limiter framework, the base schemes are defined as:

$$\psi(r)_{\text{MINBEE}} = \max\left[0, \min\left(1.5r, 0.5r + 0.5, 1.5\right)\right] \tag{2}$$

$$\psi(r)_{\text{TOPUS}} = \max\left[0, r\frac{6r+2}{(1+|r|)^3}\right] \tag{3}$$

3.1 Design of the Sensor $\omega(r)$

To capture high-frequency entropy waves without destroying shocks, $\omega(r)$ must detect the nature of the local flow field:

- **Smooth Regions ($r \approx 1$):** The gradients are continuous. The sensor must evaluate to $\omega \approx 1$. The scheme adopts TOPUS, providing third-order polynomial accuracy.
- **Severe Discontinuities/Shocks ($r \to 0$ or $r \to \infty$):** The sensor must evaluate to $\omega \approx 0$. The scheme reverts entirely to MINBEE, utilizing its strict TVD bounds to prevent spurious oscillations.

We propose the following bounded algebraic sensor:

$$\omega(r) = \left(\frac{2|r|}{r^2+1}\right)^2 \tag{4}$$

This function naturally peaks at exactly $\omega(1) = 1$ and decays asymptotically to 0 for large gradient variations, allowing a flawless dynamic transition. The final hybrid limiter becomes:

$$\psi(r)_{\text{Hybrid}} = (1 - \omega(r))\psi(r)_{\text{MINBEE}} + \omega(r)\psi(r)_{\text{TOPUS}} \tag{5}$$